(Please write your Enrollment Number) **Enrollment No.__________________**

MID TERM EXAMINATION
(Aug.-December, 2019)

Subject Code: BAS 101 **Subject: Applied Mathematics I**
Time : 1 ½ Hours **Maximum Marks :** 30
Note: Q1 is compulsory. Attempt any two questions from the rest.

Scientific Calculator Allowed

		Marks	CO Mapping
Q1	a) Prove that Eigen values of a unitary matrix are of magnitude unity. b) Prove that if the matrix A has eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, then $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$ will be the eigenvaues of A^m. c) Obtain the Fourier series expansion of $xsin\ x$ as a cosine series in (0, pi). d) Discuss the convergence of the series $\sum \frac{1}{x^n+x^{-n}}, x > 0.$	2.5 2.5 2.5 2.5	CO1 CO1 CO2 CO2
Q2	a) Solve $x^2yz = e, xy^2z^3 = e, x^3y^2z = e$ using matrices. b) Apply Cauchy's Integral Test to check the convergence of the series $\sum \frac{1}{n^2+n}$.	5 5	CO1 CO2
Q3	a) Verify Cayley-Hamilton Theorem for the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence find A^{-1}. b) Discuss the convergence of the series $1 + \frac{2}{5}x + \frac{6}{9}x^2 + \frac{14}{17}x^3 + \cdots, (x > 0)$ [3]	5 5	CO1 CO2